

Modern Agentic AI Syllabus

1. Foundations of AI and Agentic Systems

- Artificial Intelligence fundamentals
- Machine Learning vs Deep Learning vs Generative AI
- Introduction to AI Agents
- GenAI vs Agentic AI concepts
- Capabilities and limitations of AI agents

2. Programming Foundations

- Python for AI development
- Data structures and algorithms basics
- Working with APIs
- Environment management and virtual environments
- Working with Jupyter notebooks

3. Large Language Models for Agents

- Transformer architecture
- Tokenization and embeddings
- Prompt engineering fundamentals
- LLM capabilities and limitations
- Using LLM APIs

4. Core Components of Agentic AI

- LLM as reasoning engine
- Memory systems
- Planning modules
- Tool usage
- Action execution mechanisms

5. Agent Architectures

- ReAct pattern
- Plan-and-Execute agents
- Reflection agents
- Multi-agent collaboration

- Hierarchical agent systems

6. Frameworks for Building Agents

- LangChain agents
- LangGraph workflows
- CrewAI multi-agent systems
- AutoGen collaborative agents
- Comparison of agent frameworks

7. Tool Integration

- Connecting external APIs
- Custom tool creation
- Database integration
- Search tools
- Automation tools

8. Memory Systems

- Short-term memory
- Long-term memory
- Vector databases
- Knowledge retrieval
- Memory management strategies

9. Retrieval Augmented Generation for Agents

- Embeddings and semantic search
- Vector databases
- Document indexing
- RAG pipelines
- Agentic RAG architectures

10. Planning and Reasoning

- Chain-of-thought reasoning
- Tree-of-thought reasoning
- Task decomposition
- Planning algorithms
- Decision making in agents

11. Multi-Agent Systems

- Agent collaboration patterns
- Task distribution
- Agent communication
- Swarm intelligence
- Human-in-the-loop systems

12. Multimodal Agents

- Text agents
- Vision agents
- Speech agents
- Video understanding agents
- Multimodal reasoning

13. Evaluation of Agent Systems

- Performance metrics
- Task success evaluation
- Agent reliability testing
- Safety testing
- Human evaluation

14. Deployment and Infrastructure

- Containerization with Docker
- Kubernetes deployment
- Serverless AI agents
- Scalable architecture
- Monitoring agent systems

15. Responsible AI and Safety

- AI ethics
- Bias mitigation
- Security risks
- Data privacy
- Safe deployment of AI agents

16. Real-World Agentic AI Applications

- AI DevOps automation agents
- AI customer support agents
- AI research assistants
- AI coding assistants
- Autonomous workflow agents

17. Capstone Projects

- Autonomous DevOps assistant
- AI knowledge management agent
- Multi-agent research assistant
- AI troubleshooting agent
- Enterprise workflow automation agent